

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: DSIP(01CT1513)	Aim: Compute and analyze the correlation between audio tracks		
Open-ended-Problem: 03	Date:30-07-2026	Enrollment No:92400133121	

AIM:

Compute and analyze the correlation between three audio tracks:

- 1.the original song,
- 2.the karaoke (music-only) version
- 3.a completely different song.

Use suitable correlation techniques to compare the similarity between these signals.

THEORY:

WHAT IS CORRELATION?

- The correlation of two functions or signals or waveforms is defined as the measure of similarity between those signals.
- They are widely used in various applications like signal processing, image processing and pattern recognition.
- There are two types of correlations-
 1. Cross-correlation
 2. Auto-correlation

1.Cross – Correlation

- The cross-correlation between two different signals or functions or waveforms is defined as the measure of similarity or coherence between one signal and the time-delayed version of another signal.
- The cross-correlation between two different signals indicates the degree of relatedness between one signal and the time-delayed version of another signal.

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- There are also many applications of signal cross correlation in signal processing systems, especially when the signal is corrupted by another undesirable signal (noise) so that the signal estimation (detection) from a noisy signal has to be performed.

$$R_{xy}[k] = \sum_{m=-\infty}^{\infty} x[m]y[m-k], \quad R_{yx}[k] = \sum_{m=-\infty}^{\infty} y[m]x[m-k]$$

2.Auto – Correlation

- The autocorrelation function is defined as the measure of similarity or coherence between a signal and its time delayed version.
- Therefore, the autocorrelation is the correlation of a signal with itself.
- Physically, signal autocorrelation indicates how the signal energy (power) is distributed within the signal, and as such is used to measure the signal power.
- Typical applications of signal autocorrelation are in radar, sonar, satellite, and wireless communications systems.
- Devices that measure signal power using signal correlation are known as signal correlators.

$$R_{xx}[k] = \sum_{m=-\infty}^{\infty} x[m]x[m-k], \quad R_{yy}[k] = \sum_{m=-\infty}^{\infty} y[m]y[m-k]$$

****AUDIO PREPROCESSING ****

- ➡ MP3 audio files are first loaded using PyDub library.
- ➡ Audio samples are converted into NumPy arrays for numerical computation.
- ➡ Samples are further normalized.

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- ➡ Then all the audio signals are trimmed to equal length for proper analysis.
- ➡ In this case we doing cross-correlation as we want to compare three different audio signals.

Case-1: Audio Duration: 5sec

```
from pydub import AudioSegment
import numpy as np
import matplotlib.pyplot as plt
```

```
def cross_correlation(signal1, signal2):
# cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr
```

```
original = AudioSegment.from_mp3("audio1.mp3")
different = AudioSegment.from_mp3("audio2.mp3")
karaoke = AudioSegment.from_mp3("audio1_karaoke.mp3")
```

```
original = original[:5000] #5sec
different = different[:5000] #5sec
karaoke = karaoke[:5000] #5sec
```

```
original = original.set_channels(1)
different = different.set_channels(1)
karaoke = karaoke.set_channels(1)
```

```
samples_original = np.array(original.get_array_of_samples()).astype(np.float32)
samples_different = np.array(different.get_array_of_samples()).astype(np.float32)
samples_karaoke = np.array(karaoke.get_array_of_samples()).astype(np.float32)
```

```
original_norm = samples_original / np.max(np.abs(samples_original))
different_norm = samples_different / np.max(np.abs(samples_different))
karaoke_norm = samples_karaoke / np.max(np.abs(samples_karaoke))
```

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```
min_length = min(len(original_norm),len(different_norm),len(karaoke_norm))
original_norm = original_norm[:min_length]
different_norm = different_norm[:min_length]
karaoke_norm = karaoke_norm[:min_length]
```

```
original_karaoke = cross_correlation(original_norm,karaoke_norm)
original_different = cross_correlation(original_norm,different_norm)
different_karaoke = cross_correlation(different_norm,karaoke_norm)
```

```
print("Correlation Values")
print("Original - Karaoke : ",np.max(np.abs(original_karaoke)))
print("Original - Different : ",np.max(np.abs(original_different)))
print("Different - Karaoke : ",np.max(np.abs(different_karaoke)))
```

```
plt.figure(figsize=(10,10))
plt.subplot(3,1,1)
plt.plot(original_karaoke)
plt.title('Cross_corr_Original_Karaoke')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.subplot(3,1,2)
plt.plot(original_different)
plt.title('Cross_corr_Original_Different')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.subplot(3,1,3)
plt.plot(different_karaoke)
plt.title('Cross_corr_Different_Karaoke')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.tight_layout()
plt.show()
```

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Result:

Execution time:

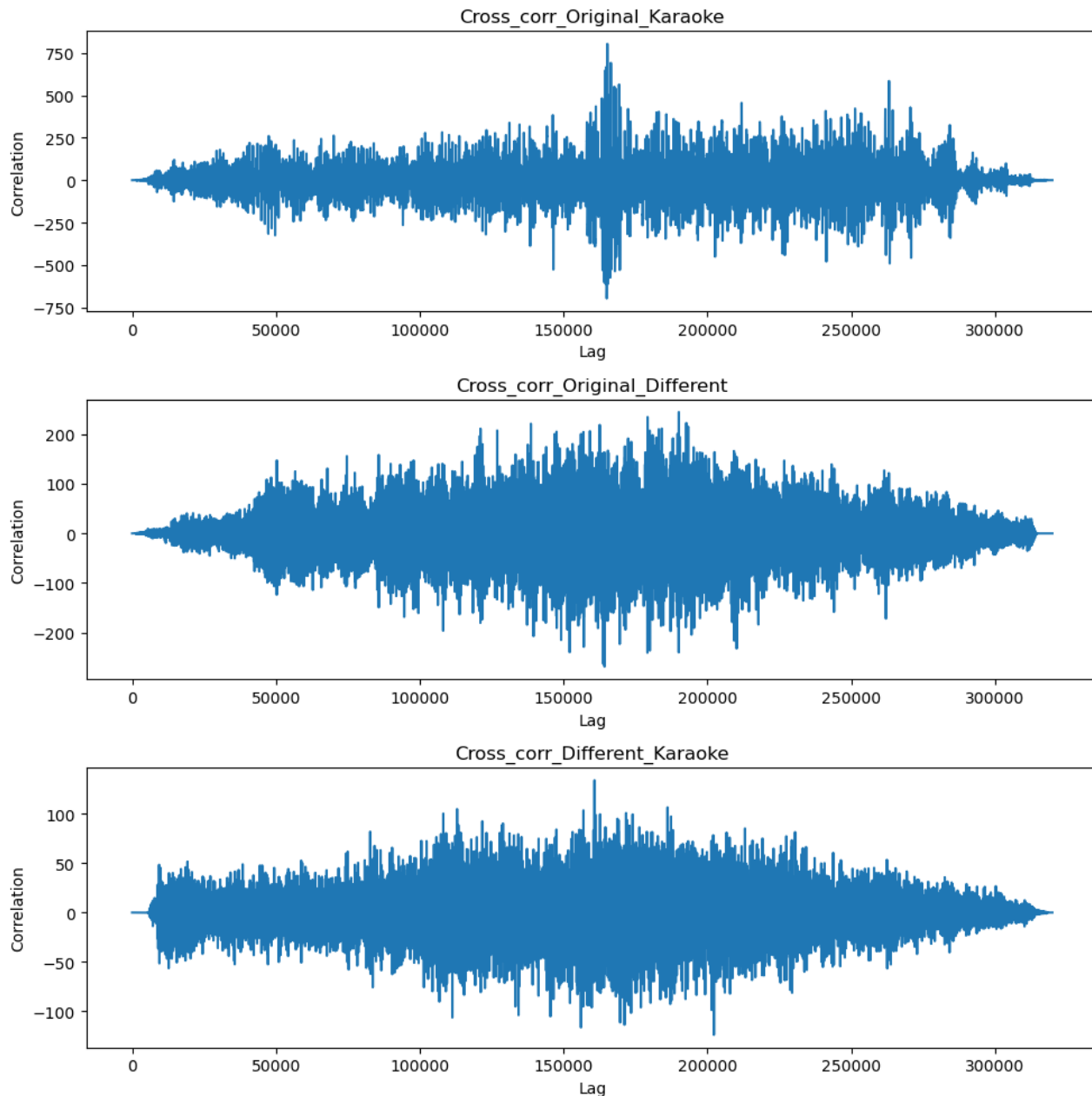
✓ 12.4s

Peak Correlation Values:

Correlation Values
Original - Karaoke : 803.4512
Original - Different : 268.5732
Different - Karaoke : 133.83295

Waveform:

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Case – 2: Audio Duration: 10sec

```
from pydub import AudioSegment
import numpy as np
import matplotlib.pyplot as plt
```

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```

def cross_correlation(signal1, signal2):
# cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr

original = AudioSegment.from_mp3("audio1.mp3")
different = AudioSegment.from_mp3("audio2.mp3")
karaoke = AudioSegment.from_mp3("audio1_karaoke.mp3")

original = original[:10000] #10sec
different = different[:10000] #10sec
karaoke = karaoke[:10000] #10sec

original = original.set_channels(1)
different = different.set_channels(1)
karaoke = karaoke.set_channels(1)

samples_original = np.array(original.get_array_of_samples()).astype(np.float32)
samples_different = np.array(different.get_array_of_samples()).astype(np.float32)
samples_karaoke = np.array(karaoke.get_array_of_samples()).astype(np.float32)

original_norm = samples_original / np.max(np.abs(samples_original))
different_norm = samples_different / np.max(np.abs(samples_different))
karaoke_norm = samples_karaoke / np.max(np.abs(samples_karaoke))

min_length = min(len(original_norm),len(different_norm),len(karaoke_norm))
original_norm = original_norm[:min_length]
different_norm = different_norm[:min_length]
karaoke_norm = karaoke_norm[:min_length]

original_karaoke = cross_correlation(original_norm,karaoke_norm)
original_different = cross_correlation(original_norm,different_norm)
different_karaoke = cross_correlation(different_norm,karaoke_norm)

print("Correlation Values")

```

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```
print("Original - Karaoke : ",np.max(np.abs(original_karaoke)))
print("Original - Different : ",np.max(np.abs(original_different)))
print("Different - Karaoke : ",np.max(np.abs(different_karaoke)))
```

```
plt.figure(figsize=(10,10))
plt.subplot(3,1,1)
plt.plot(original_karaoke)
plt.title('Cross_corr_Original_Karaoke')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.subplot(3,1,2)
plt.plot(original_different)
plt.title('Cross_corr_Original_Different')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.subplot(3,1,3)
plt.plot(different_karaoke)
plt.title('Cross_corr_Different_Karaoke')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.tight_layout()
plt.show()
```

Result

Execution time:

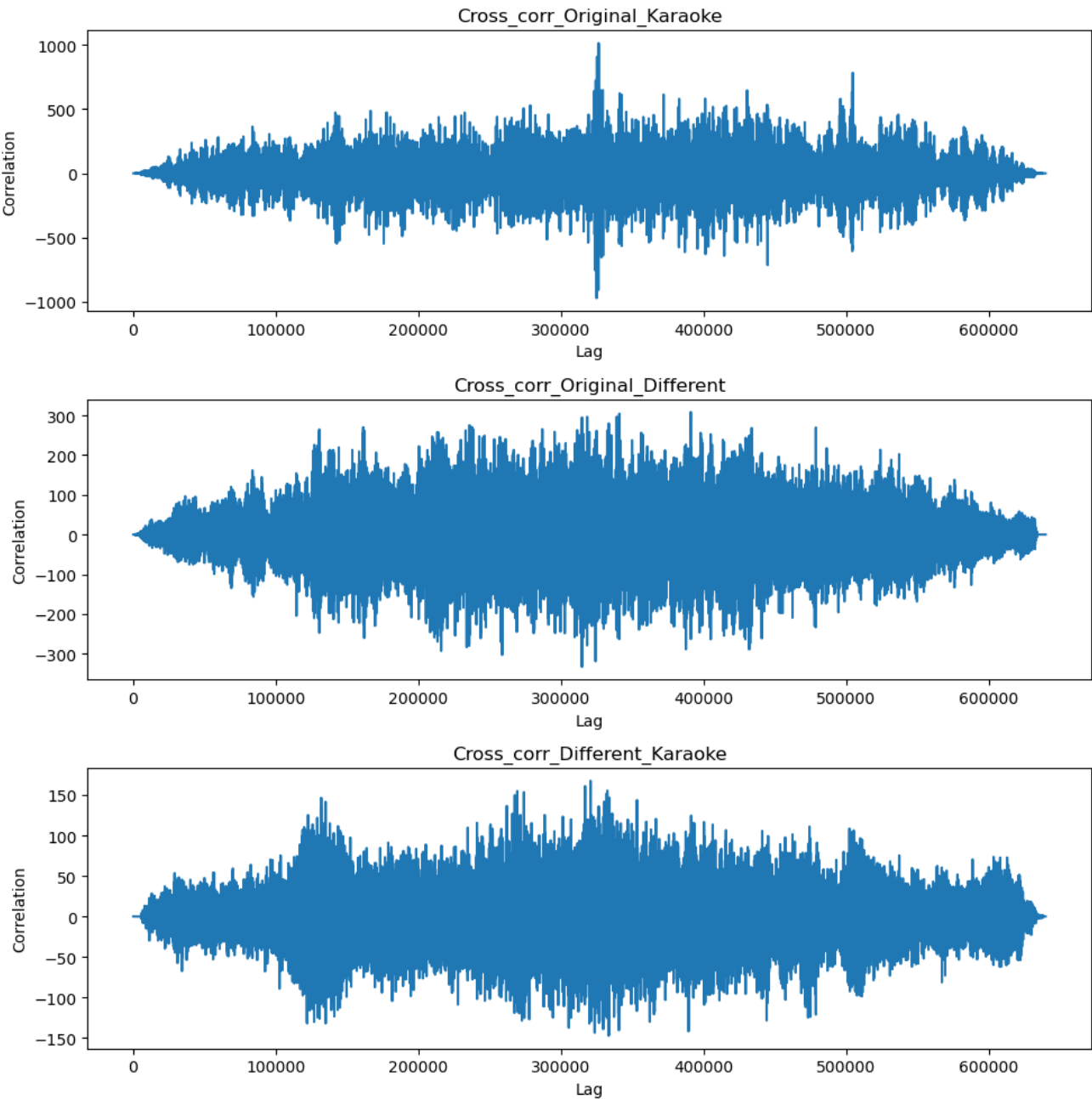
✓ 43.1s

Peak Correlation Values:

```
Correlation Values
Original - Karaoke : 1013.46405
Original - Different : 333.5112
Different - Karaoke : 167.39726
```

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Waveform:



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Case – 3: Audio Duration: 30sec

```
from pydub import AudioSegment
import numpy as np
import matplotlib.pyplot as plt
```

```
def cross_correlation(signal1, signal2):
# cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr
```

```
original = AudioSegment.from_mp3("audio1.mp3")
different = AudioSegment.from_mp3("audio2.mp3")
karaoke = AudioSegment.from_mp3("audio1_karaoke.mp3")
```

```
original = original[:30000] #30sec
different = different[:30000] #30sec
karaoke = karaoke[:30000] #30sec
```

```
original = original.set_channels(1)
different = different.set_channels(1)
karaoke = karaoke.set_channels(1)
```

```
samples_original = np.array(original.get_array_of_samples()).astype(np.float32)
samples_different = np.array(different.get_array_of_samples()).astype(np.float32)
samples_karaoke = np.array(karaoke.get_array_of_samples()).astype(np.float32)
```

```
original_norm = samples_original / np.max(np.abs(samples_original))
different_norm = samples_different / np.max(np.abs(samples_different))
karaoke_norm = samples_karaoke / np.max(np.abs(samples_karaoke))
```

```
min_length = min(len(original_norm),len(different_norm),len(karaoke_norm))
original_norm = original_norm[:min_length]
different_norm = different_norm[:min_length]
karaoke_norm = karaoke_norm[:min_length]
```

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```
original_karaoke = cross_correlation(original_norm,karaoke_norm)
original_different = cross_correlation(original_norm,different_norm)
different_karaoke = cross_correlation(different_norm,karaoke_norm)
```

```
print("Correlation Values")
print("Original - Karaoke : ",np.max(np.abs(original_karaoke)))
print("Original - Different : ",np.max(np.abs(original_different)))
print("Different - Karaoke : ",np.max(np.abs(different_karaoke)))
```

```
plt.figure(figsize=(10,10))
plt.subplot(3,1,1)
plt.plot(original_karaoke)
plt.title('Cross_corr_Original_Karaoke')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.subplot(3,1,2)
plt.plot(original_different)
plt.title('Cross_corr_Original_Different')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.subplot(3,1,3)
plt.plot(different_karaoke)
plt.title('Cross_corr_Different_Karaoke')
plt.xlabel('Lag')
plt.ylabel('Correlation')
```

```
plt.tight_layout()
plt.show()
```

Execution time:

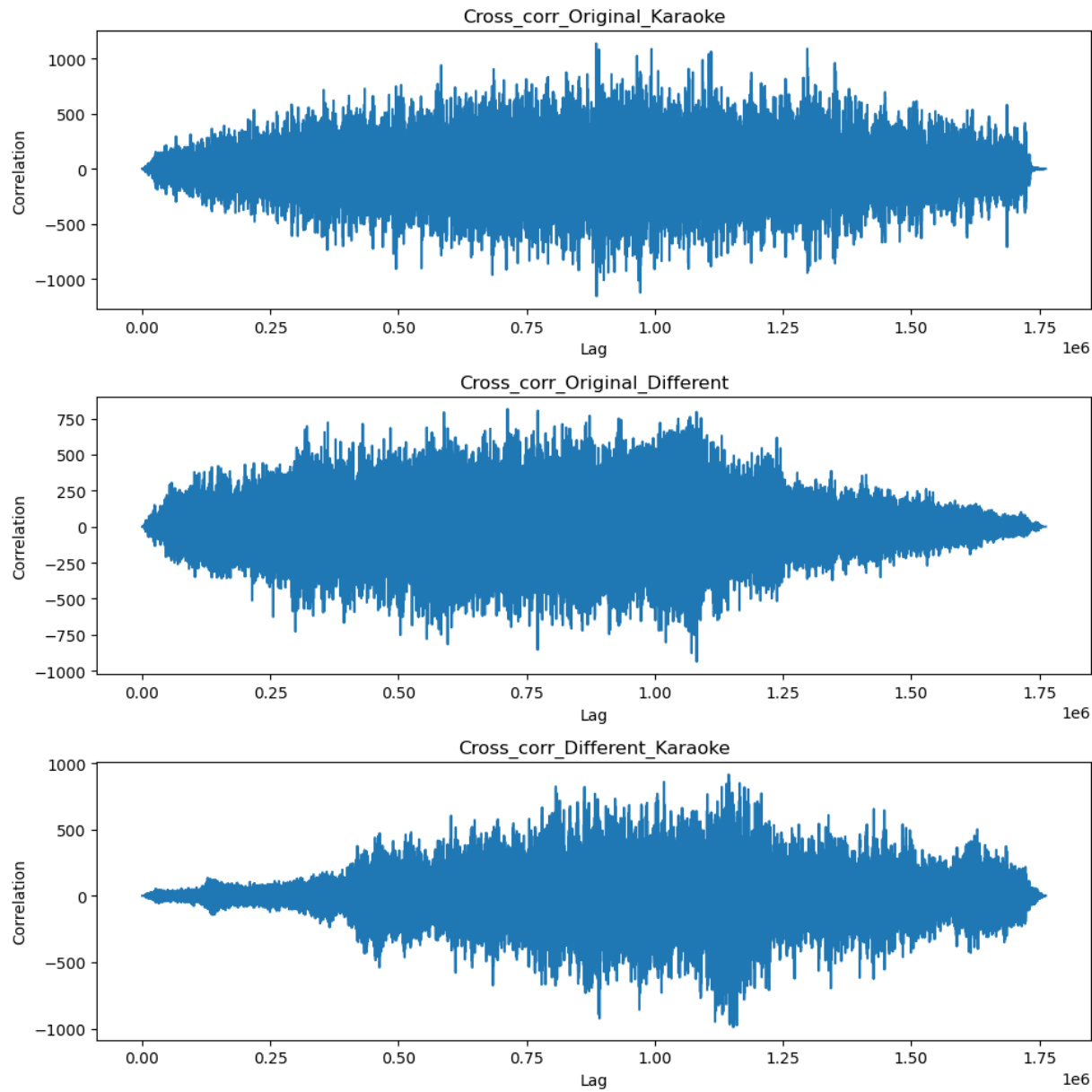
✓ 7m 34.8s

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Correlation Values:

Correlation Values
Original - Karaoke : 1157.3445
Original - Different : 937.9475
Different - Karaoke : 990.18896

Waveform:



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Observation:

Audio Duration	Original - Karaoke	Original - Different	Karaoke (Original)- Different
5sec	803.4512	268.5732	133.83295
10sec	1013.46405	333.5112	167.39726
30sec	1157.3445	937.9475	990.18896

Original Song Vs Karaoke Version

- ✓ The highest peak correlation value between original song and its karaoke version indicates the strong similarity.
- ✓ The karaoke version does not have any vocals but has same instrumental music, rhythm and timing.
- ✓ Hence, highest correlation values indicate highest similarity between signals.

Original Vs Different Song

- ✓ The graph shows significantly smaller peak compared to original-karaoke version.
- ✓ Both the audios are completely different songs, their rhythm, lyrics, melody everything is different.
- ✓ Hence, the lower correlation values indicate less similarity between the signals.

Different Song Vs Original's Karaoke Version

- ✓ Compared to all three comparisons this gives the lowest peak value and graph shows only small fluctuations.

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- ✓ The Karaoke version belongs to one song and the other audio belongs to entirely different song so naturally they do not share much common features.
- ✓ Hence, the lowest correlation values indicate these signals are completely unrelated.

****ANALYSIS****

- As we can observe the third pair (different song vs original's karaoke version) slightly show more similarity than the second pair (original vs different song) for 30 second clip.
- Cross – correlation compares two signals over multiple time lags. As the audio duration increases, the peak correlation values for all pairs increased because a larger number of samples contributed to the computation leading to increase the computation time.
- The 30second clips required approximately 7m 34.8 seconds to execute which makes the implementation expensive.
- The 10second clips improved the execution time considerably
- Using 5second clips resulted in drastic reduction of run time maintaining same qualitative conclusion.
- Hence, selecting 5 seconds clips will be a good choice as it provides proper correlation analysis and computational efficiency

Conclusion

- The objective of this experiment was successfully achieved by computing and analysing the cross – correlation between original song, its karaoke version and a different song.
- It is observed that correlation is an effective technique for comparing the audio signals.

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- The experiment shows that increasing the audio duration increases both computation time and peak correlation values because more audio samples are processed.
- By evaluating 30 second, 10second, 5second audio clips, it was determined that a 5second provided an efficient balance computational cost and meaningful analysis.
- Overall, this experiment provided practical experience in audio preprocessing, signal normalization, cross-correlation analysis, and performed optimization using python.